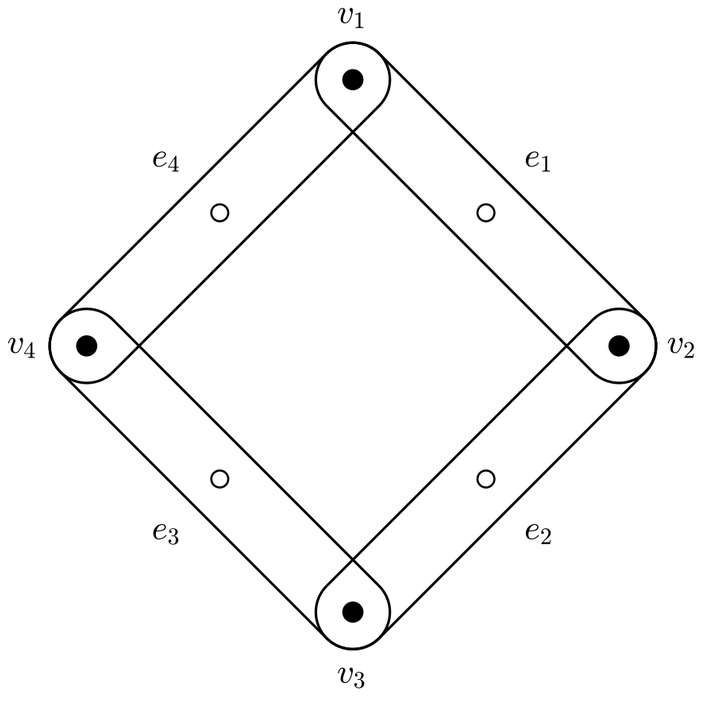
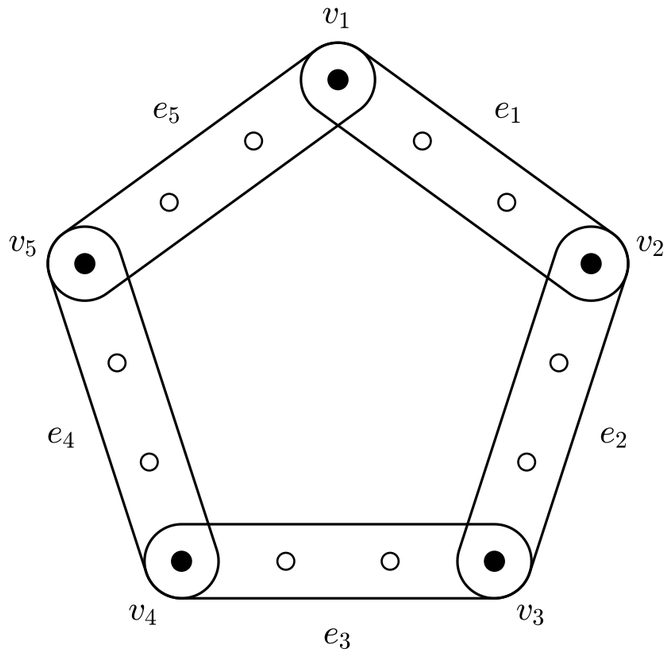
(a) $C_4^{(3)}$



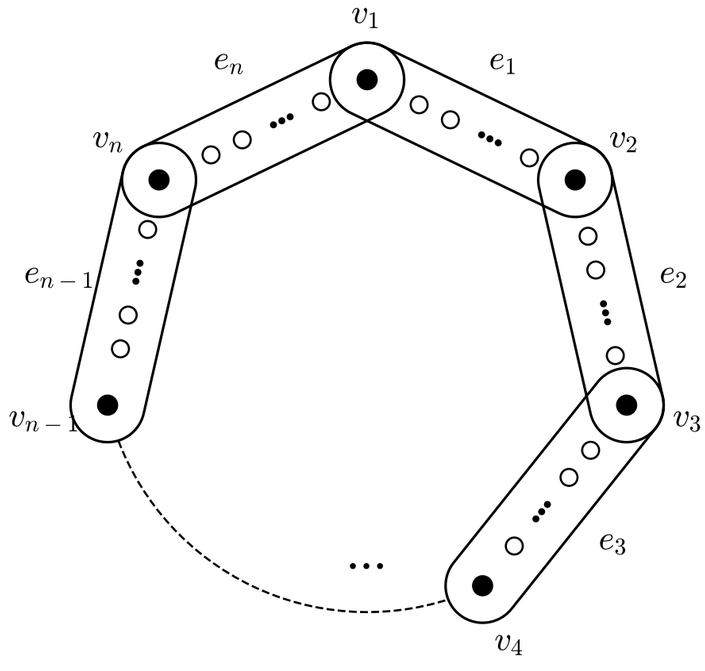
$|V| = 4 \cdot 2 = 8,\ |E| = 4$

(b) $C_5^{(4)}$



$|V| = 5 \cdot 3 = 15,\ |E| = 5$

(c) $C_n^{(k)}$: each edge has $k-2$ inner vertices



$|V| = n(k-1),\ |E| = n$

$e_i = \{v_i,\ u_{i,1}, \ldots, u_{i,k-2},\ v_{i+1}\}$ (indices mod n): consecutive edges share exactly one vertex, non-consecutive edges are disjoint